Probabilistic Modelling Coursework 2

Deadline: 30th April 2025 6PM

Instructions:

1) You need to submit a **single** PDF file
2) You are free to use any programming language to answer questions, but you **MUST** show evidence of code that was used. This can be screenshots of the code (see "Sample_Solution.pdf")
3) You will **NOT** be marked for your coding ability, but you **MUST** show evidence of code that was used.
4) Your coursework will **NOT** be marked if you do not include the proof of code used to answer the questions
5) The mark breakdown for the assessment is:

    Question 1: 20 marks

    Question 2: 25 marks

    Question 3: 35 marks

    Presentation of report: 15 marks

    Total: 95 marks

6) Your presentation will be marked down if:
    - Figures do not have the appropriate axis
    - Figures are not labelled
    - Equations are not labelled appropriately
7) Note that there is no word count limit/page limit. So, show all your working out.

By submitting my assignment:

I declare that I have personally prepared this assignment. The work is my own, carried out personally by me unless otherwise stated and has not been generated using Artificial Intelligence tools unless specified as a clearly stated approved component of the assessment brief. All sources of information, including quotations, are acknowledged by means of the appropriate citations and references. I declare that this work has not gained credit previously for another module at this or another University. I understand that plagiarism, collusion, copying another student and commissioning (which for the avoidance of doubt includes the use of essay mills and other paid for assessment writing services, as well as unattributed use of work generated by Artificial Intelligence tools) are regarded as offences against the University's Assessment Regulations and may result in formal disciplinary proceedings.

I understand that by submitting this assessment, I declare myself fit to be able to undertake the assessment and accept the outcome of the assessment as valid.

Question 1 [20 marks]

Consider a probability distribution which is an equal mixture of two Poisson distribution parameterised by $\alpha\lambda$ and $(1-\alpha)\lambda$ where $\alpha \in (0,1)$ and $\lambda > 0$ are some constants.

a) Write the PDF/PMF of the coupled distribution
   [3 marks]
b) Find the $\mathbb{E}[X]$ if X is a random variable sampled from the coupled distribution
   [10 marks]
c) Find the variance of the distribution
   [7 marks]

Question 2 [25 marks]

You are working for a car dealership which operates in over 10 countries. Your job is to compare how similar some dealerships are. The managing director is interested in comparing how similar the London dealership is to the following three:

- The New York dealership
- The Doha dealership
- The Paris dealership

You have been given the data on the number of sales per day ("London.csv", "NY.csv", "Doha.csv", and "Paris.csv"). Some dealerships are older (like London) and have much more data than others. The data has been randomly shuffled (so avoid using any time-series modelling).

a) Plot the data of each dealership, and comment on what you see
[3 marks]
b) Which dealership is the most similar to the London dealership? You may only use the methods introduced in **this module**
[10 marks]
c) Assume that for a given dealership the number of sales per day are i.i.d samples from a distribution. The distribution is the same, but the parameters are dependent on the location of the dealership.
For example:

You may assume a different Binomial model for each dealership:
The London office

$$x_i|London \sim Binomial(N_l, p_l)$$

The Doha office

$$x_i|Doha \sim Binomial(N_d, p_d)$$

The NY office

$$x_i|NY \sim Binomial(N_n, p_n)$$

The Paris office

$$x_i|Paris \sim Binomial(N_p, p_p)$$

Where all locations use the Binomial distribution but have different parameters.

You must find an **appropriate distribution** (this does not need to be the same as the example I showed above) to model the data and find

the maximum likelihood estimation of the respective parameters that depend on the location.
[12 marks]

Question 3 [35 marks]

You've been given the dataset "OldFaithful.dat" (each includes a serial number and two real values- eruptions and waiting time). Use the last two digits of your student number as the serial number of your first data point and save the next 150 points. Use the Expectation Maximisation (EM) algorithm to train a Gaussian mixture model (GMM) of the data.

a) Explain briefly the GMM model and the EM algorithm you are using.
   [4 marks]
b) Train a GMM of $K = \{2,3,4,5\}$ Gaussians and plot the resulting mixture model (in the form of Gaussians in this two-dimensional space).
   [8 marks]
c) Plot the progression in likelihood values with respect to the iterations of the EM algorithm.
   [4 marks]
d) Review methods for finding the optimal number of Gaussians in the mixture, explain the various criteria and the one you have been using.
   [4 marks]
e) Based on your results determine which of the models used best represents the data.
   [5 marks]
f) Explain which software package you have used and how it implements the algorithm described in (a) or provide the code if written by yourself.
   [10 marks]